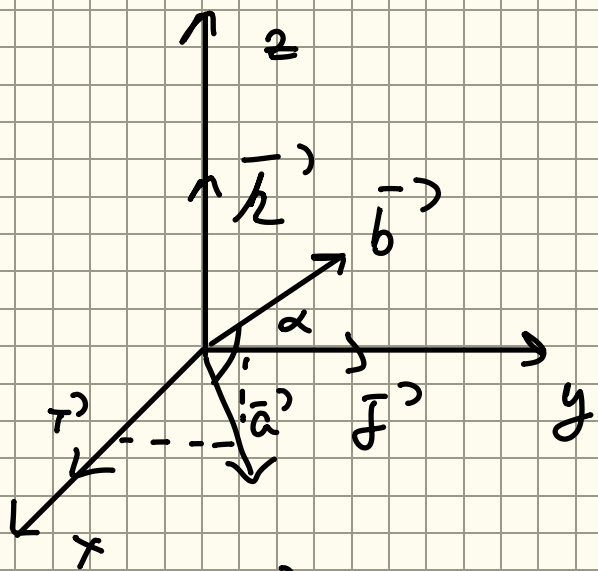




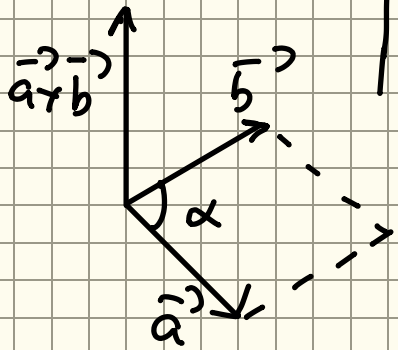
$$\Rightarrow \vec{a} = x_A \vec{i} + y_A \vec{j} + z_A \vec{k}$$

$$= (x_A, y_A, z_A)$$

$$\vec{b} = x_B \vec{i} + y_B \vec{j} + z_B \vec{k}$$

$$\vec{a} \cdot \vec{b} = x_A x_B + y_A y_B + z_A z_B = |\vec{a}| \cdot |\vec{b}| \cdot \cos \alpha$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_A & y_A & z_A \\ x_B & y_B & z_B \end{vmatrix} = \vec{i} \cdot \begin{vmatrix} y_A & z_A \\ y_B & z_B \end{vmatrix} - \vec{j} \cdot \dots$$



$$\Rightarrow |\vec{a} \times \vec{b}| = |\vec{a}| \cdot |\vec{b}| \cdot \sin \alpha$$

1.1) Decompose $\vec{v} = \vec{i} + 3\vec{j} - 5\vec{k}$ as a linear combination of $\vec{a} = \vec{i} + 2\vec{j}$, $\vec{b} = \vec{i} + 2\vec{k}$, $\vec{c} = 2\vec{i} - \vec{j} + \vec{k}$

We look for scalars: $\alpha, \beta, \gamma \in \mathbb{R}$ n.t.: $\vec{v} = \alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}$

$$= \alpha \vec{i} + 2\alpha \vec{j} + \beta \vec{i} + 2\beta \vec{k} + 2\gamma \vec{i} - \gamma \vec{j} + \gamma \vec{k}$$

$$\vec{v} = \vec{i} (\alpha + \beta + 2\gamma) + \vec{j} (2\alpha - \gamma) + \vec{k} (2\beta + \gamma)$$

$$= \underline{\vec{i}} + \underline{3\vec{j}} - \underline{5\vec{k}}$$

$$\Rightarrow \begin{cases} \alpha + \beta + 2\gamma = 1 \\ 2\alpha - \gamma = 3 \\ 2\beta + \gamma = -5 \end{cases}$$

$$2(\alpha + \beta) = -2$$

$$\Rightarrow \alpha + \beta = -1$$

$$\Rightarrow \gamma = 1$$

$$\Rightarrow \alpha = 2$$

$$\beta = -3$$

$$\Rightarrow \vec{v} = 2\vec{a} - 3\vec{b} + \vec{c}$$

$$\Rightarrow \Leftrightarrow [A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 0 & -1 & 3 \\ 0 & -2 & 1 & -5 \end{array} \right] \xrightarrow{L_2 - 2L_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & -2 & -5 & 1 \\ 0 & -2 & 1 & -5 \end{array} \right] \xrightarrow{L_3 + L_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & -2 & -5 & 1 \\ 0 & 0 & -4 & -4 \end{array} \right] \Leftrightarrow \begin{cases} \alpha + \beta + 2\gamma = 1 \\ -2\beta - 5\gamma = 1 \\ -4\gamma = -4 \end{cases}$$

$$\Rightarrow \gamma = 1$$

$$\beta = -3$$

$$\alpha = 2$$

$$\Rightarrow (\alpha, \beta, \gamma) = (2, -3, 1)$$

1.2) Let $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$, $\vec{b} = \vec{j} - 2\vec{k}$, $\vec{c} = \vec{i} - \vec{j} + 2\vec{k}$

Compute:

a) $\vec{a} + \vec{b} = ?$

c) $\vec{b} \cdot \vec{c} = ?$

e) $\widehat{(\vec{a}, \vec{b})} = ?$

b) $|\vec{b}| = ?$

d) $\vec{a} \times \vec{b} = ?$

f) $\vec{a} \cdot (\vec{b} + \vec{c}) = ?$

g) $(\vec{a} \times \vec{b}) \cdot \vec{c}$

h) The projection of $\vec{a}$ onto the vector $\vec{b}$ ($\vec{b}, m_{\vec{b}}, \vec{a}$)

a) $\vec{a} + \vec{b} = 2\vec{i} - 2\vec{j} - \vec{k}$

b) $|\vec{b}| = \sqrt{1^2 + 2^2} = \sqrt{5}$

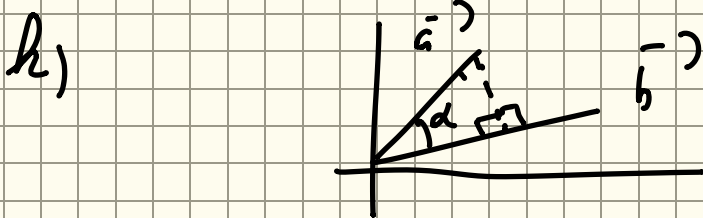
c) $\vec{b} \cdot \vec{c} = |\vec{b}| \cdot |\vec{c}| \cdot \cos \theta = -1 - 4 = -5$

d) $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 1 \\ 0 & 1 & -2 \end{vmatrix} = \vec{i} \begin{vmatrix} -3 & 1 \\ 1 & -2 \end{vmatrix} - \vec{j} \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} + \vec{k} \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix} = 5\vec{i} + 4\vec{j} + 2\vec{k}$

$$e) \cos \theta = \frac{-3 \cdot -2}{\sqrt{14} \cdot \sqrt{5}} = \frac{-5}{\sqrt{70}} \Rightarrow \theta = \arccos\left(\frac{-5}{\sqrt{70}}\right)$$

$$f) \vec{a} \cdot (\vec{b} + \vec{c}) = 2 \cdot 1 = 2$$

$$g) (\vec{c} \times \vec{b}) \cdot \vec{c} = 5 - 4 + 4 = 5$$



$$|\vec{a} \cdot \vec{b}| = |\vec{a}| \cdot |\vec{b}| \cdot |\cos \alpha| = |\sqrt{14}| \cdot \left| \frac{-5}{\sqrt{70}} \right| = \sqrt{5}$$

1.3) Consider 3 points $A(-2, 1, 0)$, $B(2, 0, 3)$, $C(-2, 0, 4)$, $D(-4, 1, 3)$. Determine:

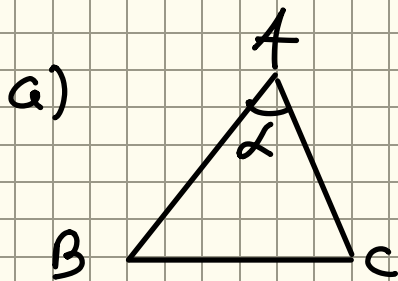
a) $A_{\triangle ABC}$

d) V_{ABCD}

b) $d(D, \text{plane } ABC)$

e) h_{ABCD} , with base $\triangle ABC$

c) $\angle(\vec{AB}, \vec{CD})$



$$A_{\triangle ABC} = \frac{1}{2} \cdot |\vec{AB}| \cdot |\vec{AC}| \cdot \sin \alpha$$

$$= \frac{1}{2} \cdot |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} = 4\vec{i} - \vec{j} + 3\vec{k}$$

$$\vec{AC} = -\vec{j} + 4\vec{k}$$

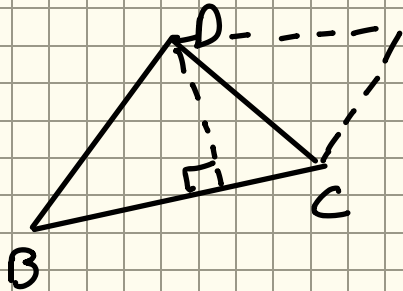
$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -1 & 3 \\ 0 & -1 & 4 \end{vmatrix}$$

$$= \vec{i} \cdot \begin{vmatrix} -1 & 3 \\ -1 & 4 \end{vmatrix} - \vec{j} \cdot \begin{vmatrix} 4 & 3 \\ 0 & 4 \end{vmatrix} + \vec{k} \cdot \begin{vmatrix} 4 & -1 \\ 0 & -1 \end{vmatrix} = -\vec{i} - 16\vec{j} - 4\vec{k}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{1 + 16^2 + 4^2} = \sqrt{17 + 256} = \sqrt{273}$$

$$A_{ABC} = \frac{\sqrt{273}}{2}$$

b)



$$A_{D,BC} = |\vec{BD} \times \vec{BC}|$$

$$A_{D,BC} = d(D, BC) \cdot |\vec{BC}|$$

$$\Rightarrow d(D, BC) = \frac{|\vec{BD} \times \vec{BC}|}{|\vec{BC}|}, \quad \vec{BD} = -6\vec{i} + \vec{j}$$

$$\vec{BC} = -4\vec{i} + \vec{k}$$

$$\vec{BD} \times \vec{BC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -6 & 1 & 0 \\ -4 & 0 & 1 \end{vmatrix} = \vec{i} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} -6 & 0 \\ -4 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} -6 & 1 \\ -4 & 0 \end{vmatrix}$$

$$= \vec{i} + 6\vec{j} + 4\vec{k}$$

$$d(D, BC) = \frac{\sqrt{1 + 36 + 16}}{\sqrt{16 + 1}} = \frac{\sqrt{53}}{\sqrt{17}}$$

$$c) \cos \alpha = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| \cdot |\vec{CD}|}$$

$$|\vec{AB}| = \sqrt{16 + 1 + 9} = \sqrt{26}$$

$$|\vec{CD}| = \sqrt{36 + 1 + 1} = \sqrt{38}$$

$$\vec{AB} \cdot \vec{CD} = -24 - 1 - 9 = -34 \Rightarrow \alpha = \arccos\left(\frac{-34}{\sqrt{26} \cdot \sqrt{38}}\right)$$

$$d) V_{ABCD} = \frac{1}{6} \cdot |(\vec{AB}, \vec{AC}, \vec{AD})| \cdot ABCD \text{ piramida}$$

$$\begin{aligned} \vec{AB} &= (1, -1, 3) \\ 4\vec{C} &= (0, -1, 4) \\ 4\vec{D} &= (-2, 0, 3) \end{aligned} \quad \Rightarrow \quad \begin{vmatrix} 1 & -1 & 3 \\ 0 & -1 & 4 \\ -2 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 0 & -1 & 9 \\ 0 & -1 & 4 \\ -2 & 0 & 3 \end{vmatrix}$$

$$= -2 \cdot (-1)^4 \cdot \begin{vmatrix} -1 & 9 \\ -1 & 4 \end{vmatrix} = -2(-4+9) = -10$$

$$V_{ABCD} = \frac{1}{6} \cdot |-10| = \frac{5}{3}$$

$$e) V_{ABCD} = \frac{d(D, ABC) \cdot A_{ABC}}{3} \Rightarrow d(D, ABC) = \frac{3 V_{ABCD}}{A_{ABC}}$$

$$= \frac{3 \cdot \frac{5}{3}}{\frac{\sqrt{293}}{2}} = \frac{10}{\sqrt{293}}$$

1.5) Determine $\lambda \in \mathbb{R}$ s.t. $\vec{a} = \vec{i} + 2\lambda\vec{j} - (\lambda-1)\vec{k}$,
 $\vec{b} = (3-\lambda)\vec{i} + \vec{j} + 2\vec{k} : a \perp b$
 $\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow 3-\lambda + 2\lambda - 2\lambda + 2 = 0$
 $\Rightarrow \lambda = 5$

1.6) Determine the angle between $\vec{u}$ and $\vec{v}$,
 if: $|\vec{u}| = 2$, $|\vec{v}| = 4$ and $2\vec{u} + \vec{v} \perp (3\vec{u} - 2\vec{v})$
 $(2\vec{u} + \vec{v}) \cdot (3\vec{u} - 2\vec{v}) = 0$
 $6|\vec{u}|^2 - 4\vec{u} \cdot \vec{v} + 3\vec{u} \cdot \vec{v} - 2|\vec{v}|^2 = 0$
 $6 \cdot 4 - \vec{u} \cdot \vec{v} - 2 \cdot 16 = 0 \Leftrightarrow \vec{u} \cdot \vec{v} = -8$

$$= 0 \quad \cos \alpha = \frac{-1}{2 \cdot 1} = -1 \Rightarrow \alpha = \pi$$